

Three Approaches to Improve Inferences Based on Survey Data Collected with Mixed-Mode Designs

Journal of Survey Statistics and Methodology (2024) 12, 814–839

Reporter: YUNG-CHUN LAN, YUE-CHEN WU



|| Outline

1. Intro
 2. Methods
 3. Simulation
 4. Replication
 5. Application
 6. Conclusion
-

Introduction

What's Mixed-mode designs?

Use two or more data collection methods (online, phone, face-to-face) within one study to boost response rates, reduce costs, and reach diverse populations

The challenge of Mixed-mode

- Mode measurement effects : social desirability bias, cognitive processing, item non-response
- Mode selection effects : nonresponse error

Literature to solve the challenge

- Developing Estimate-level Weights & Imputation
- Calibration to a Reference Mode

Context of this study

Mixed-mode is necessity

Due to COVID-19, the data collection is limited by restricted social contact

Case : Arab Barometer

- In 2020, they change FTF to mixed-mode(FTF and TEL)
- Different views in differ literatures
 - FTF enhanced rapport built between interviewers and respondents
 - no clear differences between FTF and TEL on sensitive items

How to make inferences to incorporate the potential mode effects by proposing three new approaches in this article

Testimator

Bayesian

Model Averaging

Assumption

1. Data distribution and mode

- Normal distribution
- Two modes

2. Population Means and Bias

- Unbiasedness (At least one mode is unbiased)

3. Observation and Independence

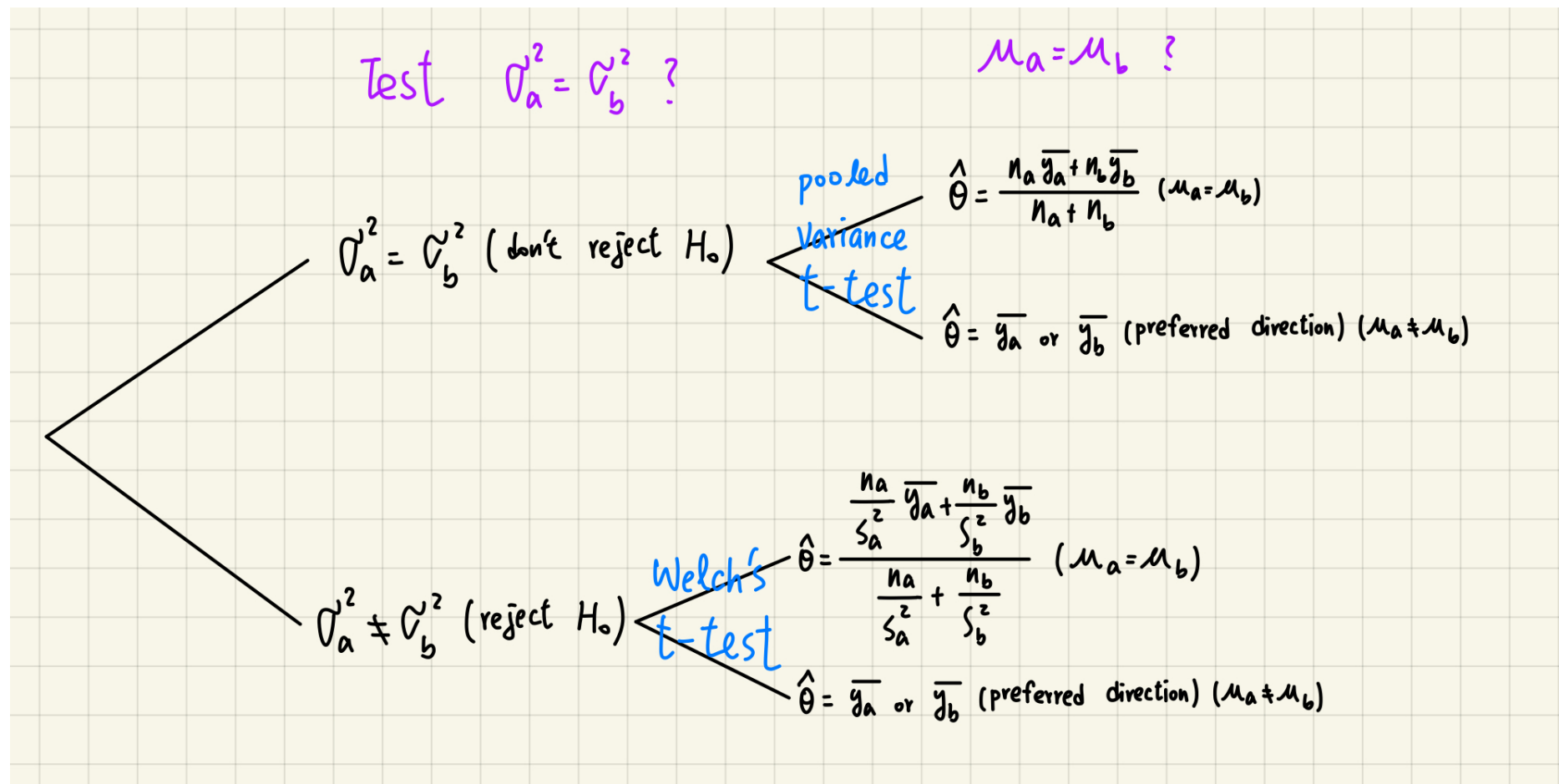
- No joint observation
- Conditional independence

4. Inference Settings

- Known Preferred Direction
- No Mode Effects
- Unknown Preferred Direction

Methods

Testimator approach- preferred direction



Testimator approach- no preferred direction

$\mu_a = \mu_b$ Same as previous consequence

$$\mu_a \neq \mu_b \quad [L_a, U_a] = \left[\bar{y}_a - t_{\frac{\gamma}{2}(n_a-1)} \frac{s_a}{\sqrt{n_a}}, \bar{y}_a + t_{\frac{\gamma}{2}(n_a-1)} \frac{s_a}{\sqrt{n_a}} \right]$$

$$[L_b, U_b] = \left[\bar{y}_b - t_{\frac{\gamma}{2}(n_b-1)} \frac{s_b}{\sqrt{n_b}}, \bar{y}_b + t_{\frac{\gamma}{2}(n_b-1)} \frac{s_b}{\sqrt{n_b}} \right]$$

$$[L, U] = [\min(L_a, L_b), \max(U_a, U_b)]$$

$$\hat{\theta} = \frac{L + U}{2}$$

The Bayesian Approach-preferred direction

consider mode A, B

$$y_{a_i} \sim N(\mu_a, \sigma_a^2) \quad y_{b_i} \sim N(\mu_b, \sigma_b^2)$$

$$\text{conjugate prior for } \mu_a, \mu_b, \sigma_a^2, \sigma_b^2: \mu_a | \sigma_a^{-2} \sim N(\beta_a, \frac{\sigma_a^2}{k_a}) \quad \sigma_a^{-2} \sim G(\frac{v_a}{2}, \frac{v_a}{2} \tau_a^2)$$

$$\mu_b | \sigma_b^{-2} \sim N(\beta_b, \frac{\sigma_b^2}{k_b}) \quad \sigma_b^{-2} \sim G(\frac{v_b}{2}, \frac{v_b}{2} \tau_b^2)$$

β_a, β_b : reflect prior belief about μ_a, μ_b

k_a, k_b : control contribution of prior information to μ_a, μ_b
(posterior mean)
 $k \uparrow$, contribution $\uparrow$

τ_a^2, τ_b^2 : prior estimate of precision

v_a, v_b : level of confidence in τ^2

Effect size

$$\hat{\eta} = \frac{\hat{\mu}_a - \hat{\mu}_b}{\sqrt{\frac{\hat{\sigma}_a^2 + \hat{\sigma}_b^2}{2}}}$$

The Bayesian Approach-preferred direction

Create cutoff value for η -hat by computing 50,75,90,95 percent quantile-based credible interval of η -hat, check if it contains 0

$$(\xi_l, \xi_u)$$

(a.) $0 \in (\xi_l, \xi_u) \Rightarrow$ assume common μ for modes ($\hat{\theta} = \mu$)

assume prior $\mu \sim N(\beta_0, \psi^2)$

$$y_{a_i} \sim N(\mu, \sigma_a^2) \quad y_{b_i} \sim N(\mu, \sigma_b^2)$$

$$\sigma_a^{-2} \sim G\left(\frac{V_a}{2}, \frac{V_a}{2} \tau_a^2\right) \quad \sigma_b^{-2} \sim G\left(\frac{V_b}{2}, \frac{V_b}{2} \tau_b^2\right)$$

$$(b.) \quad 0 < \xi_l \quad \left(\begin{array}{c} | \\ \xi_l \end{array} \begin{array}{c} | \\ \xi_u \end{array} \right) \Rightarrow \hat{\theta} = \mu_b$$

assume prior $\mu_b \sim N(\beta_b, \psi_b^2)$

$$y_{b_i} \sim N(\mu_b, \sigma_b^2)$$

$$\sigma_b^{-2} \sim G\left(\frac{V_b}{2}, \frac{V_b}{2} \tau_b^2\right)$$

$$(c.) \quad \xi_u > 0 \quad \left(\begin{array}{c} | \\ \xi_l \end{array} \begin{array}{c} | \\ \xi_u \end{array} \begin{array}{c} | \\ 0 \end{array} \right) \Rightarrow \hat{\theta} = \mu_a$$

assume prior $\mu_a \sim N(\beta_a, \psi_a^2)$

$$y_{a_i} \sim N(\mu_a, \sigma_a^2)$$

$$\sigma_a^{-2} \sim G\left(\frac{V_a}{2}, \frac{V_a}{2} \tau_a^2\right)$$

prefer
smaller
estimate
for
example

The Bayesian Approach- no preferred direction

$$0 \in (\varepsilon_l, \varepsilon_u)$$

(A.) $0 \in (\varepsilon_l, \varepsilon_u) \Rightarrow$ assume common μ for modes ($\hat{\theta} = \mu$)

assume prior $\mu \sim N(\beta_0, \psi^2)$

$$y_{a_i} \sim N(\mu, \sigma_a^2) \quad y_{b_i} \sim N(\mu, \sigma_b^2)$$

$$\sigma_a^{-2} \sim G\left(\frac{v_a}{2}, \frac{v_a}{2} \tau_a^2\right) \quad \sigma_b^{-2} \sim G\left(\frac{v_b}{2}, \frac{v_b}{2} \tau_b^2\right)$$

$$0 \notin (\varepsilon_l, \varepsilon_u)$$

R Bernoulli($p = 0.5$) random variables

If r^{th} Bernoulli draw = 0 $\rightarrow u^{(r)} = \hat{\mu}_a^{(r)}$

If r^{th} Bernoulli draw = 1 $\rightarrow u^{(r)} = \hat{\mu}_b^{(r)}$

The Model Averaging Approach-preferred direction

Model 1(μ not equal, σ not equal)

$$y_{ai} \sim N(\mu_a, \sigma_a^2), y_{bi} \sim N(\mu_b, \sigma_b^2)$$

$$\mu_a | \sigma_a^{-2} \sim N(\beta_a, \frac{\sigma_a^2}{k_a}), \mu_b | \sigma_b^{-2} \sim N(\beta_b, \frac{\sigma_b^2}{k_b})$$

$$\sigma_a^{-2} \sim \text{Gamma}(\frac{v_a}{2}, \frac{v_a}{2} \tau_a^2), \sigma_b^{-2} \sim \text{Gamma}(\frac{v_b}{2}, \frac{v_b}{2} \tau_b^2)$$

Model 2(μ equal, σ not equal)

$$y_{ai} \sim N(\mu, \sigma_a^2), y_{bi} \sim N(\mu, \sigma_b^2)$$

$$\mu \sim N(\beta_0, \varphi^2), \sigma_a^{-2} \sim \text{Gamma}(\frac{v_a}{2}, \frac{v_a}{2} \tau_a^2), \sigma_b^{-2} \sim \text{Gamma}(\frac{v_b}{2}, \frac{v_b}{2} \tau_b^2)$$

The Model Averaging Approach-preferred direction

Model 3(μ not equal, σ equal)

$$\begin{aligned}y_{ai} &\sim N(\mu_a, \sigma^2), y_{bi} \sim N(\mu_b, \sigma^2) \\ \mu_a | \sigma^{-2} &\sim N(\beta_a, \frac{\sigma^{-2}}{k_a}), \mu_b | \sigma^{-2} \sim N(\beta_b, \frac{\sigma^{-2}}{k_b}) \\ \sigma^{-2} &\sim \text{Gamma}(\frac{v}{2}, \frac{v}{2} \lambda^2)\end{aligned}$$

Model 4(μ equal, σ equal)

$$\begin{aligned}y_{ai} &\sim N(\mu, \sigma^2), y_{bi} \sim N(\mu, \sigma^2) \\ \mu &\sim N\left(\beta_0, \frac{\sigma^2}{k}\right), \sigma^{-2} \sim \text{Gamma}(\frac{v}{2}, \frac{v}{2} \lambda^2)\end{aligned}$$

The Model Averaging Approach-preferred direction

➤ The posterior probability

$$\pi(M|y_a, y_b) = \frac{p(y|M)p(M)}{\sum_m p(y|M_m)p(M_m)}$$

$$W_M = \frac{\pi(M|y_a, y_b)}{\sum \pi(M|y_a, y_b)}$$

➤ Use W_M as multinomial probability, take R draws to indicate

Model 1 or 3 (μ not equal)

$$\hat{\theta} = \begin{cases} \bar{y}_a, & \text{if Mode A is preferred} \\ \bar{y}_b, & \text{if Mode B is preferred} \end{cases}$$

Model 2 or 4 (μ equal)

$$\hat{\theta} = \frac{n_a \bar{y}_a + n_b \bar{y}_b}{n_a + n_b} \quad \hat{\theta} = \frac{\frac{n_a}{S_a^2} \bar{y}_a + \frac{n_b}{S_b^2} \bar{y}_b}{\frac{n_a}{S_a^2} + \frac{n_b}{S_b^2}}$$

The Model Averaging Approach-preferred direction

- The posterior mean

$$E(\hat{\theta}) = \frac{1}{R} \sum_{r=1}^R \hat{\theta}^{(r)}$$

- The posterior variance

$$V(\hat{\theta})$$

- $(1 - \gamma)100\%$ credible interval

$$CI = \left[E(\hat{\theta}) \pm z_{\frac{1-\gamma}{2}} \sqrt{V(\hat{\theta})} \right]$$

The Model Averaging Approach-no preferred direction

Model 1 or 3 (μ not equal)

R Bernoulli($p = 0.5$) random variables

If r^{th} Bernoulli draw = 0 $\rightarrow \hat{\theta}^{(r)} = \hat{\mu}_a^{(r)}$

If r^{th} Bernoulli draw = 1 $\rightarrow \hat{\theta}^{(r)} = \hat{\mu}_b^{(r)}$

Model 2 or 4 (μ equal)

$$\hat{\theta} = \frac{n_a \bar{y}_a + n_b \bar{y}_b}{n_a + n_b}$$

$$\hat{\theta} = \frac{\frac{n_a}{S_a^2} \bar{y}_a + \frac{n_b}{S_b^2} \bar{y}_b}{\frac{n_a}{S_a^2} + \frac{n_b}{S_b^2}}$$

Naïve approach- Naïve preferred

$$\hat{\theta} = \min \text{ or } \max (\bar{y}_a, \bar{y}_b)$$

$$\hat{\theta} = \bar{y}_a \cdot \mathbb{I}(\bar{y}_a \leq \bar{y}_b) + \bar{y}_b \cdot \mathbb{I}(\bar{y}_a \geq \bar{y}_b)$$

$$\therefore E(\hat{\theta}) = \mu_a \Phi\left(\frac{\mu_b - \mu_a}{s}\right) + \mu_b \Phi\left(\frac{\mu_a - \mu_b}{s}\right) - s \cdot \phi\left(\frac{\mu_b - \mu_a}{s}\right), \quad s = \sqrt{\frac{\sigma_a^2}{n_a} + \frac{\sigma_b^2}{n_b}}$$

$$E(\hat{\theta}^2) = \left(\mu_a + \frac{\sigma_a^2}{n_a}\right) \Phi\left(\frac{\mu_b - \mu_a}{s}\right) + \left(\mu_b + \frac{\sigma_b^2}{n_b}\right) \Phi\left(\frac{\mu_a - \mu_b}{s}\right) - (\mu_a + \mu_b) s \cdot \phi\left(\frac{\mu_b - \mu_a}{s}\right)$$

$$\text{Var}(\hat{\theta}) = E(\hat{\theta}^2) - E(\hat{\theta})^2$$

Compute C.I. with these estimates and $\mathcal{Z}$ distribution

Naïve approach- Naïve pooled

Ignore mode effects→ as single mode

$$\hat{\mu} = \bar{y}$$

$$SE(\hat{\mu}) = \frac{S}{\sqrt{n_a + n_b}}, \text{ where } S: \text{ sample standard deviation of the combined data}$$

Use t distribution with df $n_a + n_b - 1$ to construct C.I.

Simulation Design

$$\begin{bmatrix} y_{ai} \\ y_{bi} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ \mu_b \end{bmatrix}, \begin{bmatrix} 1 & \rho\sigma_b \\ \rho\sigma_b & \sigma_b^2 \end{bmatrix}\right)$$

$$effect\ size = \frac{\mu_a - \mu_b}{\sqrt{\frac{\sigma_a^2 + \sigma_b^2}{2}}}$$

- 500 samples with $n_a = n_b = 250$
- $\rho = 0.5, 0.75, 0.95$
- Simulation R = 500

- **Testimator**

$\alpha_1 = \alpha_2 = 0.05$ for both the F-test and the t-test
 $\gamma = 0.05$ for constructing if the mean is different

- **Bayesian & Model Averaging**

Prior parameters:

$$\varphi^2 = 100, \beta_a = \beta_b = \beta_0 = 0, k_a = k_b = k = 0.01,$$

$$\frac{v_a}{2} = \frac{v}{2} = \frac{v_a}{2} = 0.001, \frac{v_a}{2} \tau_a^2 = \frac{v}{2} \lambda^2 = \frac{v_b}{2} \tau_b^2 = 0.001$$

$R = 5000$ posterior simulations for each model

Simulation Design

- (1) The preferred direction is known (smaller mean is preferred)
- (2) The preferred direction is unknown.

Scenarios	μ_b	σ_b^2	Effect size
1	0	1	0
2	0	2	0
3	0.3	2	0.24
4	0.3	1	0.3
5	0.5	2	0.41
6	0.5	1	0.5
7	0.7	2	0.57
8	0.7	1	0.7
9	0.7	0.5	0.81

Source: Journal of Survey Statistics and Methodology (2024) 12, 827

Comparing the methods using:

(1) Bias:

$$bias(\hat{\theta}) = \frac{\sum \hat{\theta}}{500} - \theta$$

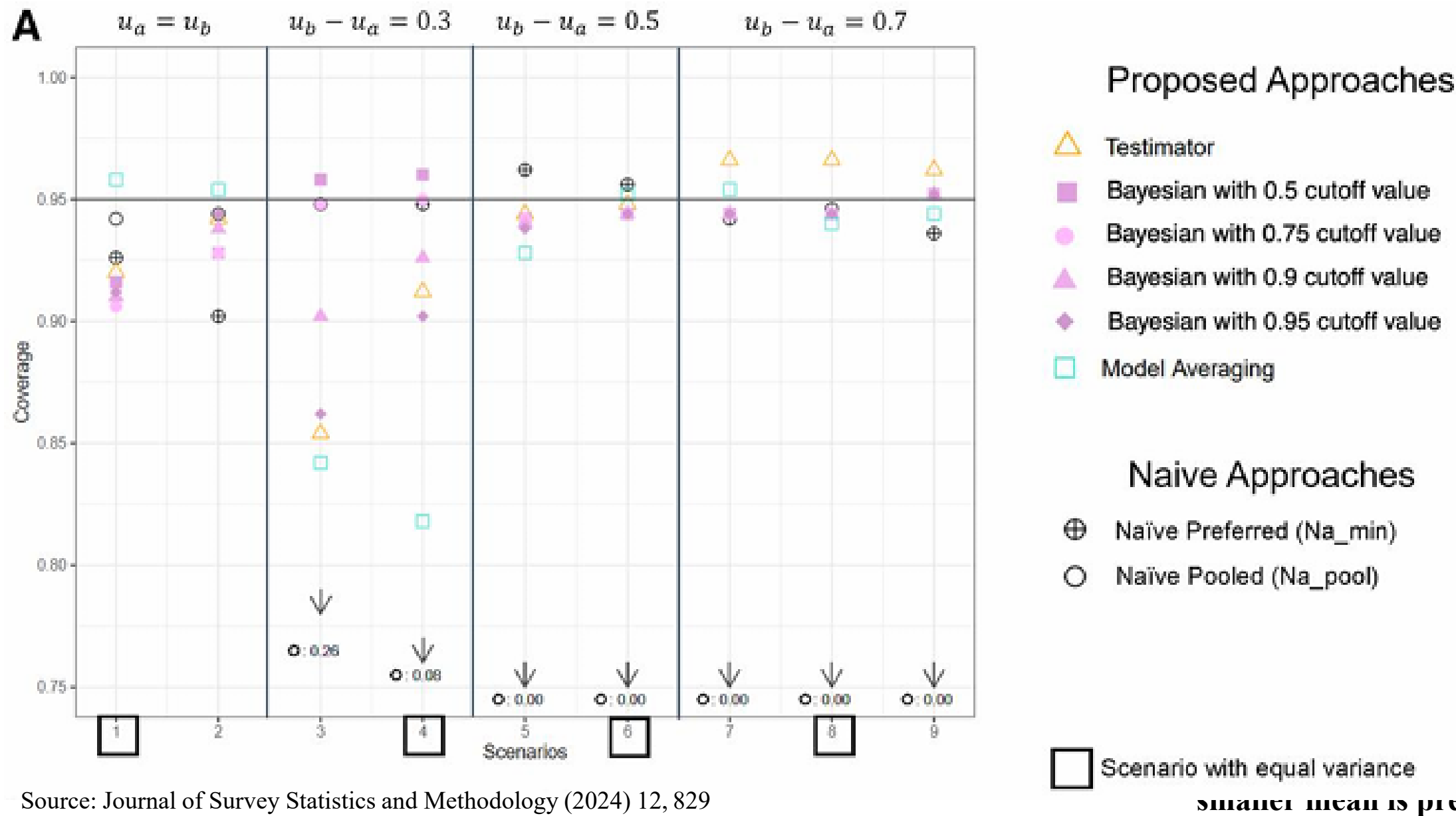
(2) RMSE:

$$RMSE(\hat{\theta}) = \sqrt{\frac{\sum (\hat{\theta} - \theta)^2}{500}}$$

(3) Coverage rate

(4) Actual interval length

(5) Expected interval length



Intro	Methods	Simulation	Replication	Application	Conclusion
-------	---------	------------	-------------	-------------	------------

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.000	0.044	0.046	0.910
$u_b = 0, \sigma_b^2 = 2$	0.000	0.058	0.942	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 2$	0.240	0.764	0.236	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.052	0.004	0.874	0.070
$u_b = 0.5, \sigma_b^2 = 2$	0.410	0.994	0.006	0.000	0.000
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.054	0.000	0.946	0.000
$u_b = 0.7, \sigma_b^2 = 2$	0.570	1.000	0.000	0.000	0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.034	0.000	0.966	0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	1.000	0.000	0.000	0.000

Probability of the Model Being Selected in the Testimator Approach ($\rho=0.95, n=500$)

Source: Journal of Survey Statistics and Methodology (2024) 12, 830

smaller mean is preferred

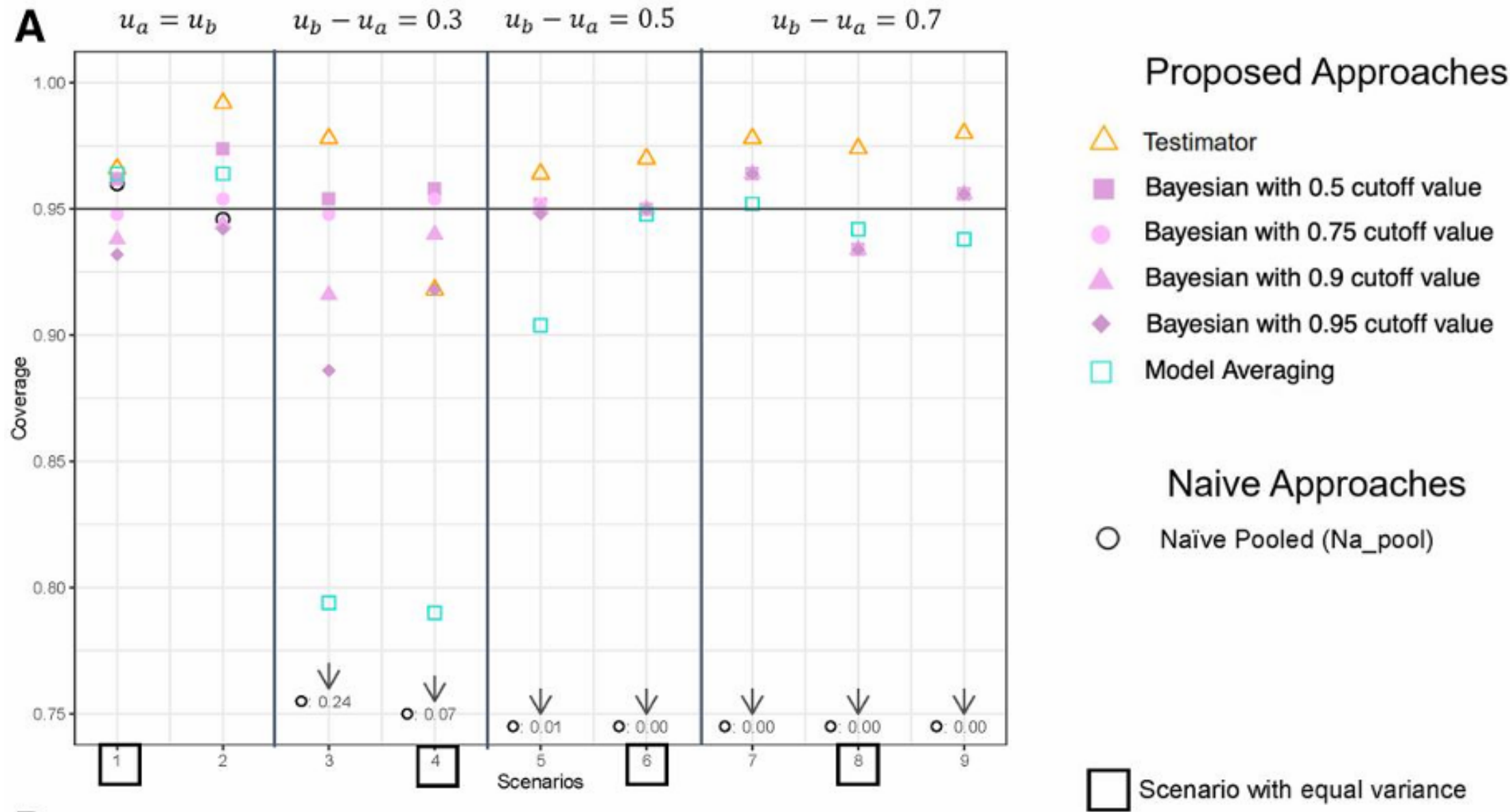
Intro	Methods	Simulation	Replication	Application	Conclusion
-------	---------	------------	-------------	-------------	------------

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.000	0.001	0.022	0.977
$u_b = 0, \sigma_b^2 = 2$	0.000	0.018	0.911	0.002	0.070
$u_b = 0.3, \sigma_b^2 = 2$	0.240	0.345	0.576	0.028	0.051
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.001	0.000	0.595	0.404
$u_b = 0.5, \sigma_b^2 = 2$	0.410	0.830	0.079	0.081	0.010
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.002	0.000	0.989	0.009
$u_b = 0.7, \sigma_b^2 = 2$	0.570	0.900	0.003	0.098	0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.001	0.000	0.999	0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	0.896	0.000	0.104	0.000

Average weight in the Model Averaging approach ($\rho = 0.95, n = 500$)

Source: Journal of Survey Statistics and Methodology (2024) 12, 831

smaller mean is preferred



Intro	Methods	Simulation	Replication	Application	Conclusion
-------	---------	------------	-------------	-------------	------------

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.002	0.036	0.048	0.914
$u_b = 0, \sigma_b^2 = 2$	0.000	0.040	0.960	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 2$	0.240	1.000	0.000	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.052	0.000	0.948	0.000
$u_b = 0.5, \sigma_b^2 = 2$	0.410	1.000	0.000	0.000	0.000
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.062	0.000	0.938	0.000
$u_b = 0.7, \sigma_b^2 = 2$	0.570	1.000	0.000	0.000	0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.042	0.000	0.958	0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	1.000	0.000	0.000	0.000

Probability of the Model Being Selected in the Testimator Approach ($\rho=0.95,n=5000$)

Source: Appendix of *Three Approaches to Improve Inferences Based on Survey Data Collected with Mixed-Mode Designs*, 61

Increase sample size to 5000 (prefer smaller mean)

Intro	Methods	Simulation	Replication	Application	Conclusion
-------	---------	------------	-------------	-------------	------------

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.000	0.000	0.002	0.998
$u_b = 0, \sigma_b^2 = 2$	0.000	0.005	0.995	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 2$	0.240	1.000	0.000	0.000	0.000
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.000	0.000	1.000	0.000
$u_b = 0.5, \sigma_b^2 = 2$	0.410	1.000	0.000	0.000	0.000
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.002	0.000	0.998	0.000
$u_b = 0.7, \sigma_b^2 = 2$	0.570	1.000	0.000	0.000	0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.000	0.000	1.000	0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	1.000	0.000	0.000	0.000

Average weight in the Model Averaging approach ($\rho = 0.95, n = 5000$)

Source: Appendix of *Three Approaches to Improve Inferences Based on Survey Data Collected with Mixed-Mode Designs*, 62

Increase sample size to 5000 (prefer smaller mean)

Method	Bias	RMSE	Coverage	Expected Len.	Actual Len.
Scenario 3: $u_b = 0.3, \sigma_b^2 = 2$					
Te	-0.001	0.02	0.942	0.081	0.079
B50	-0.001	0.021	0.958	0.077	0.078
B75	-0.001	0.021	0.958	0.077	0.078
B90	-0.001	0.021	0.958	0.077	0.078
B95	-0.001	0.021	0.958	0.077	0.078
MA	-0.001	0.02	0.948	0.079	0.078

Performance in normal outcomes when smaller estimates are preferred ($\rho = 0.95, n = 5000$)

Source: Appendix of *Three Approaches to Improve Inferences Based on Survey Data Collected with Mixed-Mode Designs*, 63

Increase sample size to 5000 (prefer smaller mean)

Replication: Data generation

The original paper does not explicitly specify the data-generating process.

We generate two correlated samples of size $n = 250$ from the following bivariate normal distribution:

$$\begin{bmatrix} y_{ai} \\ y_{bi} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ \mu_b \end{bmatrix}, \begin{bmatrix} 1 & \rho\sigma_b \\ \rho\sigma_b & \sigma_b^2 \end{bmatrix}\right)$$

- y_{ai} in the first dataset is used as the simulated data for Mode A
- y_{bi} in the second dataset is used as the simulated data for Mode B

$n = 500$, $\rho = 0.95$, and the case where a smaller mean is preferred

Replication: Posterior Distribution of Bayes Approach & Model Averaging

Different means model:

The posterior parameters of the different means model are not described in the paper, we derive them assuming a single sample for simplicity.

Common means model:

We use Gibbs sampling to generate posterior draws.

$$\text{let } T = \sigma^{-2}$$

$$f(y|\mu, t) \propto t^{\frac{1}{2}} \exp\left(-\frac{t}{2}(y-\mu)^2\right)$$

$$f(\mu|t) \propto t^{\frac{1}{2}} \exp\left(-\frac{k}{2}(\mu-\beta)^2\right) \quad (\text{prior})$$

$$f(t) \propto t^{\frac{\nu}{2}-1} \exp\left[-\frac{t}{2}\nu\tau^2\right] \quad (\text{prior})$$

$$\begin{aligned} f(\mu, t|y) &= P(y|\mu, t) \cdot P(\mu, t) \\ &= P(y|\mu, t) \cdot P(\mu|t) \cdot P(t) \\ &\propto t^{\frac{\nu}{2}} \exp\left(-\frac{t}{2}((y-\mu)^2 + \underbrace{k(\mu-\beta)^2}_{(x)} + \nu\tau^2)\right) \end{aligned}$$

① posterior of μ

$$(x) = (k+1)\left(\mu - \frac{y+k\beta}{k+1}\right)^2 + \frac{k}{k+1}(y-\beta)^2$$

$$\therefore f(\mu, t|y) \propto t^{\frac{\nu}{2}} \cdot \exp\left(-\frac{t}{2}\left(\frac{k}{k+1}(y-\beta)^2 + \nu\tau^2\right)\right) \cdot \exp\left(-\frac{t}{2}\left((k+1)\left(\mu - \frac{y+k\beta}{k+1}\right)^2\right)\right)$$

$$\Rightarrow f(\mu|t, y) \propto \exp\left(-\frac{t}{2}\left((k+1)\left(\mu - \frac{y+k\beta}{k+1}\right)^2\right)\right) \Rightarrow \mu|t, y \sim N\left(\frac{y+k\beta}{k+1}, \frac{1}{(k+1)t}\right)$$

② posterior of $\sigma^{-2}(T)$

$$f(\mu, t|y) \propto t^{\frac{\nu}{2}} \cdot \exp\left(-\frac{t}{2}\left(\frac{k}{k+1}(y-\beta)^2 + \nu\tau^2\right)\right) \cdot \exp\left(-\frac{t}{2}\left((k+1)\left(\mu - \frac{y+k\beta}{k+1}\right)^2\right)\right)$$

$$f(T|y) = \int f(\mu, t|y) d\mu \propto t^{\frac{\nu}{2}} \cdot \exp\left(-\frac{t}{2}\left(\frac{k}{k+1}(y-\beta)^2 + \nu\tau^2\right)\right) \cdot \frac{1}{\sqrt{T(k+1)}} \propto t^{\frac{\nu+1}{2}-1} \cdot \exp\left(-\frac{t}{2}\left(\frac{k}{k+1}(y-\beta)^2 + \nu\tau^2\right)\right)$$

$$\therefore T|y \sim \text{Gamma}\left(\frac{\nu+1}{2}, \frac{\frac{k}{k+1}(y-\beta)^2 + \nu\tau^2}{2}\right)$$

Structure of our code

The screenshot displays the RStudio environment with several open scripts and a console window showing the execution of R code. The code defines a function for generating data across nine scenarios, each with different parameters for μ_B and σ^2_B .

Code Snippet:

```

1 library(MASS)
2
3 # ----- function
377 # ----- main
392 ## ----- scenario 1 (mu_B=0, sigma2_B=1)
416 ## ----- scenario 2 (mu_B=0, sigma2_B=2)
440 ## ----- scenario 3 (mu_B=0.3, sigma2_B=2)
464 ## ----- scenario 4 (mu_B=0.3, sigma2_B=1)
488 ## ----- scenario 5 (mu_B=0.5, sigma2_B=2)
512 ## ----- scenario 6 (mu_B=0.5, sigma2_B=1)
536 ## ----- scenario 7 (mu_B=0.7, sigma2_B=2)
560 ## ----- scenario 8 (mu_B=0.7, sigma2_B=1)
584 ## ----- scenario 9 (mu_B=0.7, sigma2_B=0.5)

```

Environment Panel:

- Data:**
 - `fit`: List of 12
 - `sims_1` to `sims_9`: Large list (500 elements, 2.2 MB)
 - `sims_10` to `sims_18`: Large list (500 elements, 2.2 MB)
 - `tmp`: List of 12
 - `w_mat`: num [1:500, 1:4] 0.183 0.203 0.999 0.453 1 ...
- Values:**
 - `est_vec`: num [1:500] 0.00417 0.06812 -0.01634 -0.00933 -0.0324...
 - `lower_vec`: num [1:500] -0.1146 -0.0478 -0.1483 -0.1254 -0.1527 ...
 - `r`: 500L
 - `R`: 500L
 - `theta_true`: 0
 - `upper_vec`: num [1:500] 0.1229 0.1841 0.1156 0.1067 0.0878 ...
- Functions:**
 - `bayes_common_mean`: function (yA, yB, R_post = 5000, burn_in = 1000, m0...
 - `bayes_effect_size`: function (yA, yB, R_post = 5000, m0_a = 0, k0_a = 0...
 - `draw_mu_sigma2`: function (post_par, R_post = 5000)
 - `log_marginal_M1`: function (yA, yB, m0_a = 0, k0_a = 0.01, a0_a = 0.0...
 - `log_marginal_M2`: function (yA, yB, beta0_mu = 0, phi2_mu = 100, a0_a...
 - `log_marginal_M3`: function (yA, yB, m0_a = 0, k0_a = 0.01, m0_b = 0, ...
 - `log_marginal_M4`: function (yA, yB, m0 = 0, k0 = 0.01, a0 = 0.001, b0...

The status bar at the bottom indicates the current execution point: 607:1 # scenario 9 (mu_B=0.7, sigma2_B=0.5) :

Proposed Method (Original Paper Results / Our Results):

Method	Bias	RMSE	Coverage	Expected Len.	Actual Len.
Scenario 4: $u_b = 0.3, \sigma_b^2 = 1$. Effect size = 0.30					
Na_min	0.004 / 0.004	0.061 / 0.062	0.948 / 0.936	0.242 / 0.245	0.246 / 0.247
Na_pool	0.151 / 0.152	0.158 / 0.158	0.084 / 0.072	0.169 / 0.169	0.177 / 0.178
Te	0.008 / 0.010	0.072 / 0.074	0.912 / 0.884	0.285 / 0.289	0.244 / 0.242
B50	-0.002 / 0.004	0.061 / 0.063	0.960 / 0.938	0.240 / 0.245	0.248 / 0.249
B75	-0.001 / 0.005	0.062 / 0.064	0.950 / 0.944	0.245 / 0.251	0.247 / 0.248
B90	0.001 / 0.007	0.066 / 0.068	0.926 / 0.922	0.266 / 0.266	0.244 / 0.246
B95	0.004 / 0.010	0.071 / 0.074	0.902 / 0.884	0.282 / 0.287	0.242 / 0.242
MA	0.044 / 0.049	0.096 / 0.010	0.818 / 0.820	0.310 / 0.341	0.276 / 0.279

Model performance in different scenarios

Source: Appendix of *Three Approaches to Improve Inferences Based on Survey Data Collected with Mixed-Mode Designs*, 44

Proposed Method (Original Paper Results / Our Results):

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.000 / 0.000	0.044 / 0.052	0.046 / 0.052	0.910 / 0.896
$u_b = 0, \sigma_b^2 = 2$	0.000	0.058 / 0.056	0.942 / 0.944	0.000 / 0.000	0.000 / 0.000
$u_b = 0.3, \sigma_b^2 = 2$	0.240	0.764 / 0.802	0.236 / 0.198	0.000 / 0.000	0.000 / 0.000
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.052 / 0.050	0.004 / 0.006	0.874 / 0.854	0.070 / 0.090
$u_b = 0.5, \sigma_b^2 = 2$	0.410	0.994 / 0.994	0.006 / 0.006	0.000 / 0.000	0.000 / 0.000
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.054 / 0.044	0.000 / 0.000	0.946 / 0.956	0.000 / 0.000
$u_b = 0.7, \sigma_b^2 = 2$	0.570	1.000 / 1.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.034 / 0.040	0.000 / 0.000	0.966 / 0.960	0.000 / 0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	1.000 / 1.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000

Model selection proportion of Testimator

Proposed Method (Original Paper Results / Our Results):

Scenario	Effect size	M1 (D–D)	M2 (S–D)	M3 (D–S)	M4 (S–S)
$u_b = 0, \sigma_b^2 = 1$	0.000	0.000 / 0.000	0.001 / 0.001	0.022 / 0.020	0.977 / 0.979
$u_b = 0, \sigma_b^2 = 2$	0.000	0.018 / 0.019	0.911 / 0.912	0.002 / 0.002	0.070 / 0.068
$u_b = 0.3, \sigma_b^2 = 2$	0.240	0.345 / 0.366	0.576 / 0.556	0.028 / 0.031	0.051 / 0.047
$u_b = 0.3, \sigma_b^2 = 1$	0.300	0.001 / 0.001	0.000 / 0.001	0.595 / 0.594	0.404 / 0.404
$u_b = 0.5, \sigma_b^2 = 2$	0.410	0.830 / 0.809	0.079 / 0.092	0.081 / 0.090	0.010 / 0.009
$u_b = 0.5, \sigma_b^2 = 1$	0.500	0.002 / 0.001	0.000 / 0.000	0.989 / 0.983	0.009 / 0.016
$u_b = 0.7, \sigma_b^2 = 2$	0.570	0.900 / 0.908	0.003 / 0.001	0.098 / 0.091	0.000 / 0.000
$u_b = 0.7, \sigma_b^2 = 1$	0.700	0.001 / 0.001	0.000 / 0.000	0.999 / 0.999	0.000 / 0.000
$u_b = 0.7, \sigma_b^2 = 0.5$	0.810	0.896 / 0.919	0.000 / 0.000	0.104 / 0.081	0.000 / 0.000

Average model weights of Model Averaging

Application: Arab Barometer Wave 6 Jordan Experiment

➤ 2,531 respondents

FTF: 1,193

FTF(5 mins) + TEL: 1,338

(a) security performance

(b) keeping prices down

(c) response to COVID-19

(d) overall performance

(e) education system

(f) healthcare system

1 = “Completely satisfied”, 2 = “Satisfied”

3 = “Dissatisfied”, 4 = “Completely dissatisfied”

1 = “Very good”, 2 = “Good”

3 = “Bad”, 4 = “Very bad”

$$gov_sat = 25 - \sum_{i=1}^6 Q_i$$

➤ Consider two scenarios:

1. **Smaller estimate** is believed to be closer to the truth
2. The preferred direction is **unknown**, as a sensitivity analysis.

➤ Analysis process:

- (a) Survey weights, weighted means, weighted variances, and sampling variances
- (b) Design effect and **Effective sample size** for FTF and TEL.
- (c) Use the effective sample size and the design-based statistics $(\bar{y}_w, S_w^2)$ as inputs
- (d) Estimate the sampling variances for both modes using linearization, which approximates complex survey estimates with a Taylor expansion.

Results

Method	Estimate	Interval	Interval Len.
Testimator	7.521	7.197, 7.844	0.647
Bayesian (50%, 75%, 95%)	7.519	7.193, 7.849	0.656
Model Averaging	7.521	7.197, 7.851	0.654
Pool data	8.295	8.058, 8.531	0.473
FTF	7.521	7.200, 7.841	0.642
TEL	9.088	8.825, 9.351	0.526

Results when smaller estimates are preferred

Source: Journal of Survey Statistics and Methodology (2024) 12, 835

Results

Method	Estimate	Interval	Interval Len.
Testimator	8.275	7.197, 9.352	2.155
Bayesian (50%, 75%, 95%)	8.293	7.254, 9.307	2.053
Model Averaging	8.297	7.246, 9.304	2.058
Pool data	8.295	8.058, 8.531	0.473
FTF	7.521	7.200, 7.841	0.642
TEL	9.088	8.825, 9.351	0.526

Results when there is no information about preferred direction

Source: Journal of Survey Statistics and Methodology (2024) 12, 835

Core concepts

Testimator

- 2 stage hypothesis testing

Bayesian Approach

- CI of posterior effect size
- Mode effect
- Selects the model

Model Averaging

- Combine 4 models through their posterior weights
- Perform inference by averaging across models

Performance

- All methods have limited sensitivity to small mode effects when the sample size is small
- All methods are more robust than the Naïve pooled approaches.
- Increasing the sample size improves performance for all methods.